MIL-C-82630(OS)
27 June 1975
Superseding
XWS-5566
15 June 1966
XWS-6303
1 December 1967

MILITARY SPECIFICATION

CABLE, ELECTRIC, 65-CONDUCTOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers requirements for the procurement of synthetic rubber-jacketed cable assemblies embodying 65 conductors. These cables are distinguished by their size, type of insulation, features of construction, and the general conditions of their service use.

1.2 Classification. Cables covered by this specification are of two types, as follows (see 6.2.1):

(a) Type I. 65-conductor, extra flexible; for shipboard use.

(b) Type II. 65-conductor, extra flexible; for deep depth use.

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the Issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-915	Cable, Electrical, Special Purpose; General Specification for
MIL-Y-1140	Yarn, Cord, Sleeving, Cloth and Tape-Glass
MIL-R-6855	Rubber, Synthetic, Sheets, Strips Molded or Extruded Shapes
MIL-W-16878	Wire, Electrical, Insulation, Nigh Temperature
MIL-T-18404	Torpedoes, Environmental Requirements, General Specification for
MIL-C-19787	Cable, Electrical, Torpedo, 65-Conductor (for Torpedo Control Electrical Setting)

STANDARDS

Military

MIL-STD-417	Rubber Compositions, Vulcanized, General Purpose, Solid (Symbols and Tests)

Federal

Federal Test Method Standard No. 228	Cable and Wire, Insulated, Methods of Testing

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

3 REQUIREMENTS

3.1 Description. The multiconductor cables covered by this specification are extra flexible, insulated, concentric control cables having an abrasive-resistant jacket with the external and wire bundle diameters held to close tolerances. The Type I cable is suitable for continuous service on board submarines while the Type II cable is suitable for continuous service at submarine operating depth pressures.

3,2 General. Electric cable shall be of the design shown herein and shall conform to the requirements of this specification and the contract or order.

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of electric cable is required. The sample shall be manufactured using the methods and procedures proposed for the production lot. This sample will be tested as specified in 4,4.2 and is for the purpose of determining, prior to starting production, that the manufacturer's production process will produce cables that will meet the requirements of this specification.

3.2.2 Manufactured length. Unless otherwise specified in the contract or order, the manufactured length shall consist of any one single length of cable to which the impervious jacket was applied in one length,

3.2.3 Conflicts. In the event of conflict between this specification and documents invoked herein, the requirements of this specification shall apply.

3.3 Design and construction

3.3.1 Materials. All materials used in construction of the cable shall be of a quality and form which will assure compliance with the requirements of this specification and applicable drawings.

3.3.2 Conductors. All conductors shall conform to MIL-W-16878, Type E, Teflon insulated. The conductors shell be stranded, and the number, shielding, and size shall be in accordance with Table I for Type I cables and Table II for Type II cables. The wire insulation shall be color coded, but only to the extent that each size, including the shielded conductors, shall be of a different color. Individual conductors shall lie flat in their respective lay and not cross with conductors in that or other lays.

3.3.3 Cable wire bundle. The cable shall embody 65-conductors , as specified in Tables I and II and Figures 1 and 2, and in the contract or order. See 6.2.1.

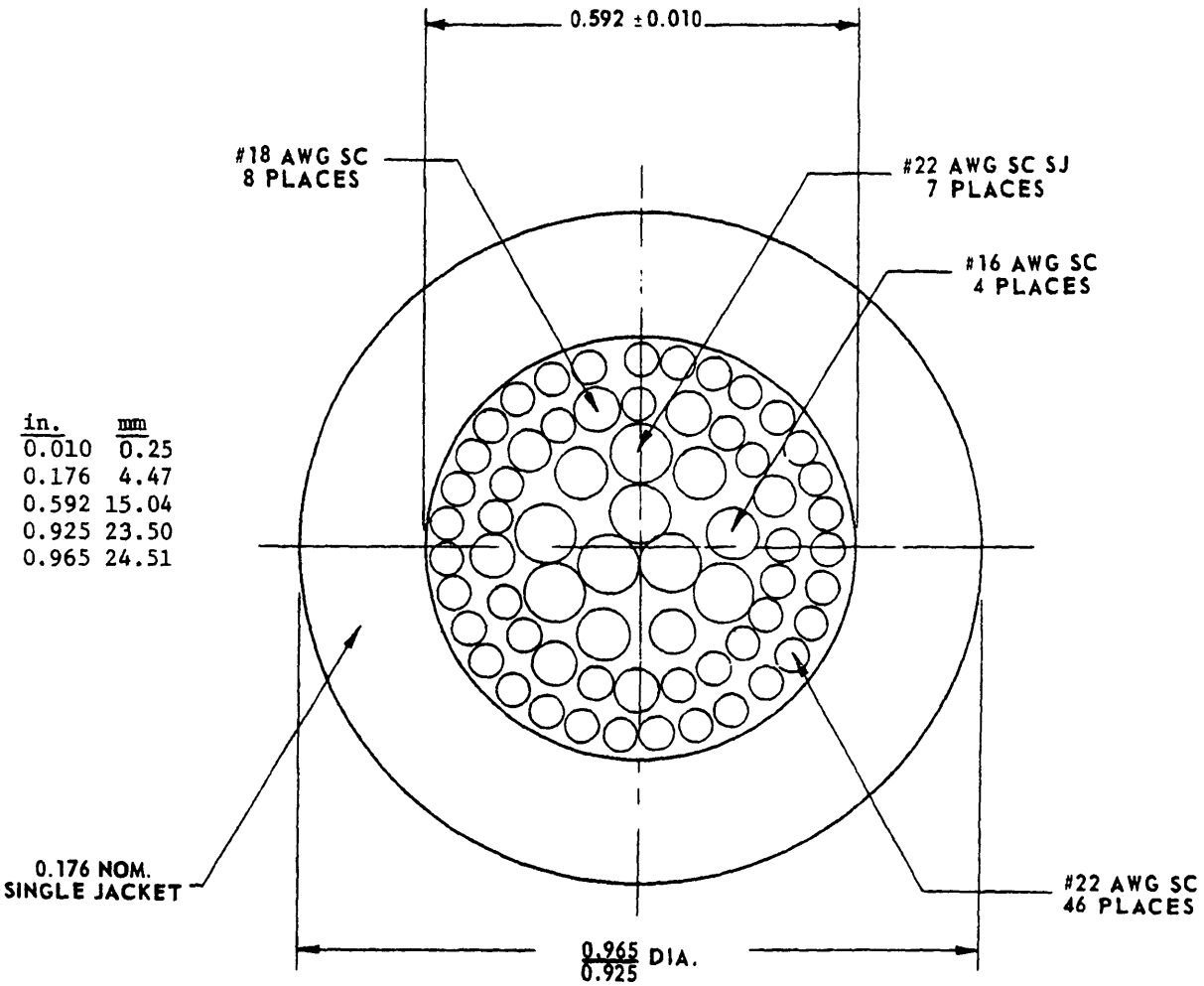
3.3.4 Cable lay. The cable shall have a concentric contrahelical lay with a pitch of 4 to 6 inches (100 to 150 mm) in the outer layer and decreasing layer by layer toward the core as necessary to provide cable flexibility. See 6.3.

3.3.5 Cable sheath. A sheath of tough resilient , abrasion resistant, polychloroprene shall be applied concentrically over the conductor assembly. The sheath shall be homogeneous in character, free from porosity, and of a quality typical of best manufacturing procedures and shall conform to MIL-R-6855, Class 2, Grade 60. In addition, the material used shall be free of residual release or other agents and be suitable for vulcanizing with neoprene in accordance with MIL-STD-417, Grade SC-420.

TABLE I

Type Conductor	Number of Conductors	AWG	Approximate Diameter		Number of Strands
			inch	mm	
Single conductor, individually shielded and jacketed	7	22	0.084	2.13	19
Single conductor	4	16	0.083	2.11	19
Single conductor	8	18	0.074	1.88	19
Single conductor	46	22	0.054	1.37	19

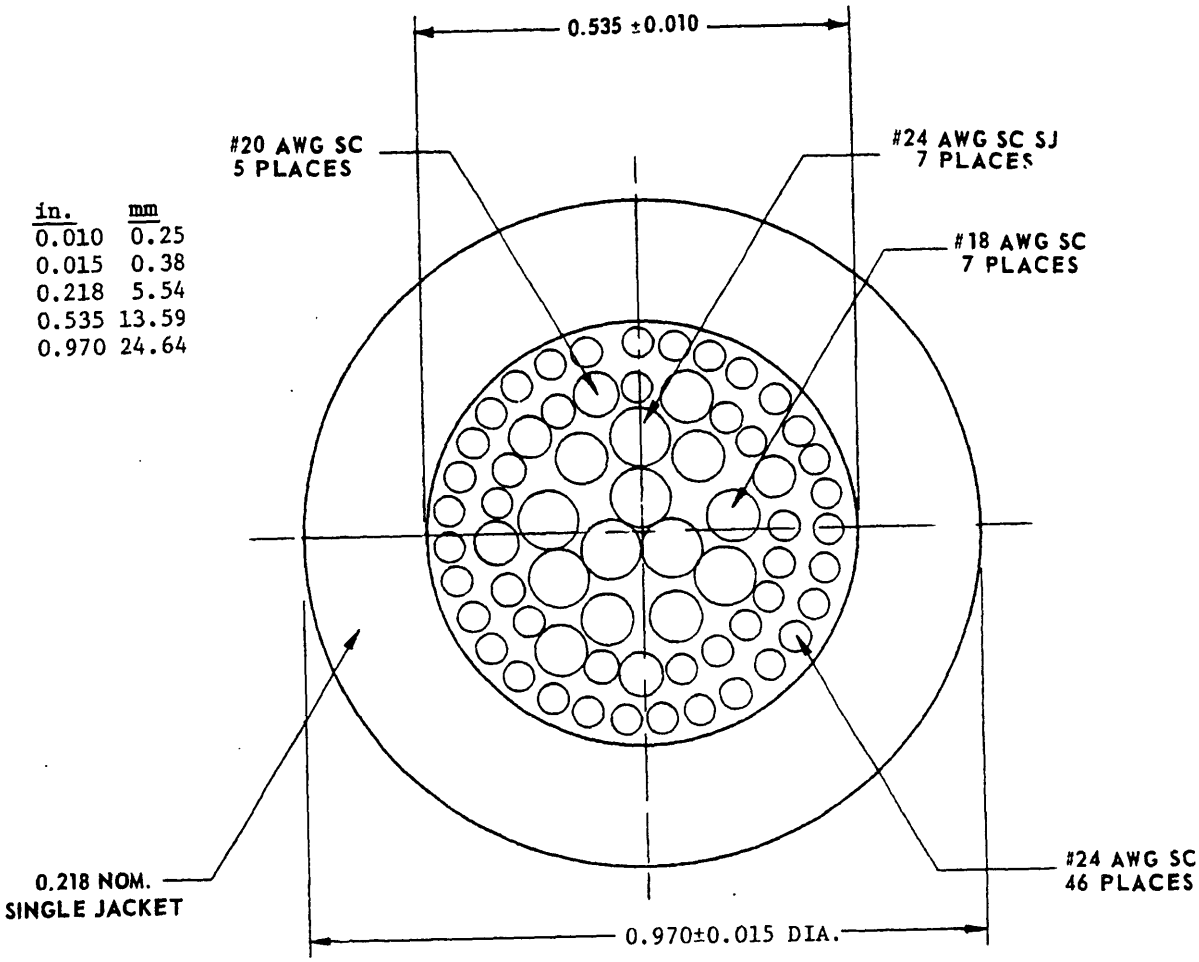
Source: https://assist.dla.mil -- Downloaded: 20260-23T07:18Z
Check the source to verify that this is the current version before use.



Note: Dimensions are shown in inches.

Figure 1. Dimensions and Tolerances - Type I Cable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.



Note: Dimensions are shown in inches.

Figure 2. Dimensions and Tolerances - Type II Cable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

TABLE II

Type Conductor	Number of Conductors	AWG	Approximate Diameter		Number of Strands
			inch	mm	
Single conductor, individually shielded and jacketed	7	24	0.082	2.08	19
Single conductor	5	20	0.062	1.57	19
Single conductor	7	18	0.074	1.88	19
Single conductor	46	24	0.048	1.22	19

3.3.6 Filler. Glass fiber filler in accordance with MIL-Y-1140 shall be used as required to make a compact circular cross section. A center filler core is acceptable.

3.3.7 End seals. Unless otherwise specified, the end seals of the cable shall be as specified in MIL-C-915.

3.3.8 Dimensions and tolerances. The overall dimensions of the cable and the jacket wall thicknesses shall conform to dimensions shown in Figures 1 and 2.

3.3.9 Wire splices. A cable length shall Include no wire splices, broken wires, broken strands of wire, or other discontinuities.

3.4 Electrical requirements. The cables shall successfully meet the following electrical requirements:

3.4.1 Dielectric strength. All lengths of completed cable shall withstand, without failure, a 5-second application of a 1000-volt ac test potential, applied to each conductor sequentially with all adjacent conductors and shields at ground potential.

3.4.2 Insulation resistance. Each length of cable shall show an insulation resistance of at least 100 megohoms for 1000 feet with 500 to 750 volts dc applied to each conductor sequentially with all other conductors and shields at ground potential.

3.4.3 Capacitance. The capacitance of each shielded lead in the cables with respect to its shield shall be not less than 55 pf nor more than 90 pf per foot (300 pf per metre), when measured at a frequency of 1 kilohertz (kHz)=20 Hz.

3.5 Mechanical requirements. The cables shall be capable of meeting the following mechanical requirements:

3.5.1 Cold temperature bend. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when the completed cables are subjected to the cold temperature 90-degree bend test of 4.5.7.

3.5.2 Cable strength. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when cables are subjected to the strength test of 4.5.8.

3.5.3 Torsion. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when completed cables are subjected to the torsion test of 4.5.9.

3.6 Environmental requirements. The cables shall be capable of meething the other requirements of this specification after being subjected co the fallowing environmental conditions:

3.6.1 Moisture penetration of cable jackets. The cable sheath, conductor insulation, and conductors shall show no evidence of moisture penetration when completed cables are subjetted co the following conditions:

(a) Type I. Immersion in salt water for 30 16-hour periods with 8 hours between immersions.

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:18Z
Check the source to verify that this is the current version before use.

(b) Type II. Exposure to 1000 psi (7MPa) of hydrostatic pressure for 90 minutes.

3.6,2 Temperature and humidity. The cable sheath, conductor insulation, and conductors shall show no evidence of failure when completed cables are subjected to the temperature and humidity test of 4.5.10.3.

3.6.3 Temperature and humidity (storage mode). The cable sheath, conductor insulation, and conductors shall show no evidence of failure when completed cables are subjected to the temperature and humidity (storage mode) test of 4.5.10,4.

3.6.4 Tropical environment. The cable sheath, conductor insulation, and conductors shall show no evidence of being fungus nutrients when the completed cable is exposed for 28 days to 86° F (30° C), 90 percent relative humidity, and fungus.

3.7 Workmanship. The workmanship shall be first-class in every respect and uniform throughout each lot of cable offered for delivery. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements relative to workmanship insofar as not specifically covered by applicable specifications.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such Inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. Preproduction inspection. (See 4.4,2)

2. Lot acceptance inspection. (See 4.4.2)

4.3 Sampling

4.3.1 Lot. A lot shall be defined as follows, based on footage ordered:

Order	Lot
Up to 10,000 feet (3km)	Total order
10,000 to 50,000 feet (3 to 15km)	10,000 feet (3km)
Over 50,000 feet (15 km)	25,000 feet (7.5km)

4.3.2 Number of samples

4.3.2.1 Preproduction. Unless otherwise specified herein or in the contract or order (see 6.2), a 100-foot (30-metre) preproduction sample of cable shall be manufactured by the contractor. The cable shall be manufactured using the same procedures and methods and at the same plant location proposed by the contractor for the execution of the contract. The sample shall be tested for purposes of determining compliance with the requirements of the contract or order and the applicable specifications. Further production of the item by the contractor prior to approval of the procuring activity or completion of inspection and tests on the preproduction sample shall be at the contractor’s risk.

4.3.2.2 Lot acceptance. Unless otherwise specified herein or in the contract or order (see 6.2), sampling for lot acceptance shall be conducted in accordance with MIL-C-915.

4.4 Inspection and tests,

4.4.1 Test conditions. Unless otherwise specified, tests and examinations required by this specification shall be conducted under any combination of conditions within the ranges following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

(a) Temperature: 20 degrees C to 30 degrees C.

(b) Relative Humidity: Room ambient to 95 percent.

(c) Barometric Pressure: 24 to 31 inches of mercury,

4.4.2 Preproduction inspection and lot acceptance inspection. Preproduction inspection and lot acceptance inspection shall consist of the examinations and tests shown in Table III performed on the samples selected under 4.3.2.1 and 4.3.2.2. When specified in the contract or purchase order, certified inspection data covering the results of preproduction inspections shall be forwarded to the procuring activity. See 6.2.2.

4.4.3 Failures

4.4.3.1 Preproduction samples. The failure of a preproduction sample to comply with any provision of this specification shall be cause for denial of production approval. The disposition of the sample and proposed corrective action shall be in accordance with the contract or purchase order. (See 6.2)

4.4.3.2 Lot acceptance sample. The failure of a sample to comply with any provision of this specification shall be cause for rejection of the lot represented by the sample. The disposition of the sample and proposed corrective action shall be in accordance with the contract or purchase order. (See 6.2)

TABLE III

PREPRODUCTION AND LOT ACCEPTANCE INSPECTIONS

Type Test	Test Title	Requirement	Method
Preproduction	Materials	3.2, 3.3.1	4.5.1
Preproduction/ Acceptance	Visual and Dimensional Examination	3.2, 3.3 3.7	4.5.1.1 4.6
Preproduction	Cable Sheath	3.3.5	4.5.1, 4.5.2
Preproduction/ Acceptance	Dielectric Strength	3.4.1	4.5.3
Preproduction/ Acceptance	Insulation Resist-	3.4.2	4.5.4
Preproduction	Cable Lay	3.3.4	4.5.5
Preproduction	Capacitance	3.4.3	4.5.6
↓	Cold Temperature Bend	3.5.1	4.5.7
↓	Cable Strength	3.5.2	4.5.8
↓	Torsion	3.5.3	4.5.9
↓	Salt Water Immersion (Type I only)	3.6.1(a)	4.5.10.1
↓	Hydrostatic Pressure (Type II only)	3.6.1(b)	4.5.10.2
↓	Temperature and Humidity	3.6.2	4.5.10.3
↓	Temperature and Humidity (storage mode)	3.6.3	4.5.10.4
Preproduction	Tropical Environment (Fungus)	3.6.4	4.5.11

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

4.5 Test methods

4.5.1 General. The methods employed for testing component materials and cable performance shall *be* as specified in MIL-C-19787, except as modified or supplemented herein.

4.5.1.1 Visual and dimensional inspection. The sample length of cable selected under 4.3.2 shall be examined visually and dimensionally for compliance with the requirements of this specification and other applicable specific ations. In addition, each manufactured length of cable shall be examined visually for surface flaws such as pin holes and pushbacks in the sheath and shall be cut at any such defects.

4.5.2 Cable sheath. One sheath specimen shall be taken from each of the cable samples chosen for insulation testing, for physical tests to determine compliance with MIL-R-6855 and the requirements of 3.3.5.

4.5.3 Dielectric strength. A length of completed cable shall be tested dry, without immersion in water, in accordance with Federal Test Method Standard No. 228 (Method 6111) to determine conformance with 3.4.1.

4.5.4 Insulation resistance. A length of completed cable shall be tested dry, without immersion in water, in accordance with Federal Test Method Standard No. 228 (Method 6031) to determine conformance with 3.4.2.

4.5.5 Cable lay. The sample chosen for dielectric testing shall be examined to determine compliance with the requirements of 3.3.4. This examination is to be done after the completion of the dielectric test.

4.5.6 Capacitance. The samples used shall be prepared in accordance with 4.5.6.1 and shall be 20 feet (6 metres) in length. The measurement technique employed shall give results that are reproducible to three significant figures.

4.5.6.1 Preparation for tests. Prepare each 20-foot (6 metre) minimum length of cable as follows (see Figure 3):

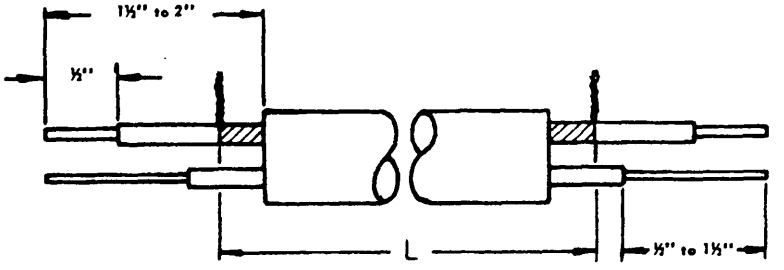
(a) Strip back cable sheathing 1 1/2 to 2 inches (40 to 50 mm) on each end of sample for connections.

(b) Comb out shielding on each of the seven shielded conductors to within approximately 1/2 inch (12 mm) of the sheathing.

(c) Strip approximately 1/2 inch (12 mm) of insulation from each shielded conductor.

NOTE: For computation purposes, the length of the cable shall be considered as the distance between the undisturbed ends of the shielding.

4.5.6.2 Test the prepared length of cable to determine conformance with 3.4.3.

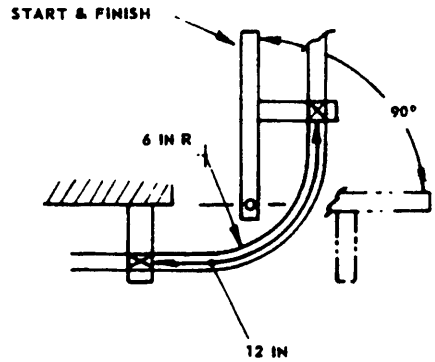


in.	mm
1/2	13
1-1/2	38
2	51

Figure 3. Preparation of Sample for Capacitance Test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

4.5.7 Cold temperature (90-degree bend test). While the cable is mounted in a suitable fixture, a 90-degree bend test shall be performed at -40 ±3 degrees (-40°±2°C), The cable assembly and test fixture shall be conditioned at the test temperature for at least 12 hours prior to initiation of the test.



in.	mm
6	152
12	305

Figure 4. Test Setup - 90-Degree Bend, Low Temperature.

The cable shall be mounted and clamped at two points on the test fixture such that a 90-degree bend with a radius not to exceed 6 inches (152mm) shall be formed within a 12-inch (305-mm) length of cable (see Figure 4). The cable shall be rididly secured to the underside of a fixed support lying in a horizontal plane at one clamp position, and rigidly secured to a pivoting arm which shall be so mounted as to oscillate from a vertical position to a horizontal position. When in its horizontal position, its axis shall form an extension of the plane of the fixed support, and the cable shall be attached to the underside of the arm. The attachment of the cable to the oscillating arm shall be made when the arm is in the vertical position. The pivot axis of the oscillating arm shall intersect the longitudinal axis of the fixed support and shall be equidistant from the two clamping points. The fixture shall be so designed that the section of cable between the clamp positions shall not touch any portion of the mechanism or cold box during the test.

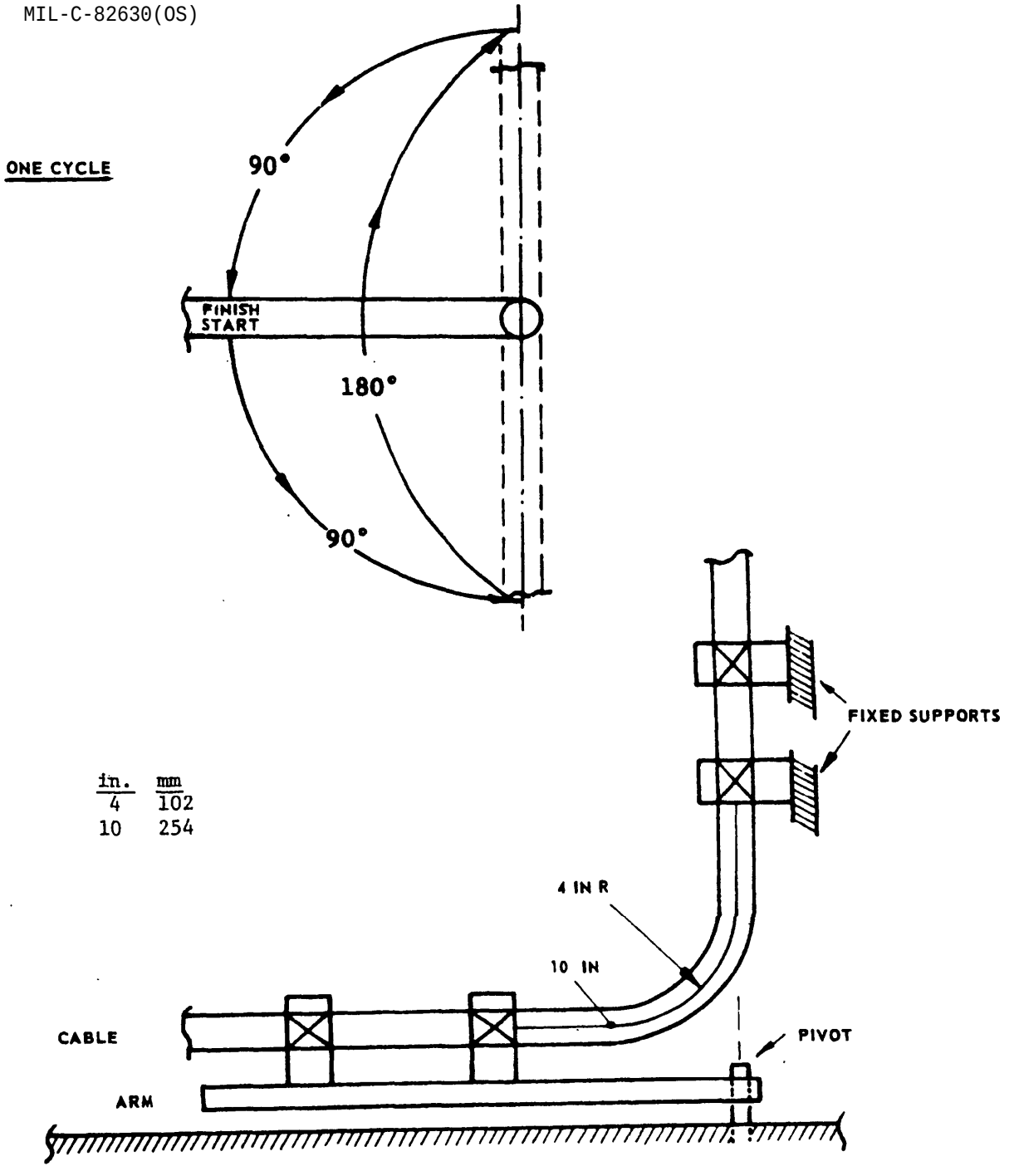
One cycle of test shall consist of an oscillation of the pivot arm from the vertical position to the horizontal and back to the vertical. The test shall be conducted at a rate of one cycle per minute with each half cycle to be accomplished within 6 seconds. The delay between cycles may occur at the end of either half cycle or may be divided to occur at the end of each half cycle.

Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded in the event of failure of any conductor. The test shall consist of 500 cycles of bending. At the conclusion of this test, the cable shall be subjected to the test of 4.5.4.

4.5.8 Cable strength. The cable shall be securely clamped at two points along its length 15 inches apart. The equipment shall be oriented such that the axis of the 15-inch (380-mm) length of cable is in a horizontal plane. At a point midway between the clamps, a weight of 300 pounds (1335 N) shall be suspended from the cable for a period of at least 60 seconds, not including the application and removal periods. The shoe or weight support shall be no more than 1-1/2 inches (38mm) wide and of such design as to spread the load over the 1-1/2-inch (38mm) length of cable. The cable sheath may be protected or padded to spread the load and prevent damage to the cable sheath. The load shall be applied and released a total of 20 times over a period not to exceed 2 hours. At the conclusion of the test, all conductors in the cable shall be tested for continuity and the cable shall be subjected to the test of 4.5.4.

4.5.9 Torsion test. While it is mounted in a suitable test fixture, a torsion test shall be performed on the cable assembly. The cable shall be mounted and clamped at two points on the test fixture such that a 90-degree bend with a radius not to exceed 4 inches (102mm) shall be formed within a 10-inch (254-mm) length of cable (see Figure 5). The cable shall be rigidly secured to a fixed support at one clamp position, and rigidly secured to a pivot arm at the second clamp position. The pivot axis of the arm shall be an extension of the centerline of the cable shich is clamped to the fixed support.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.



Torsion Test Setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

One cycle of torsion shall consist of a 90-degree rotation in one direction, a 180-degree rotation in the reverse direction, and a 90-degree rotation in the original direction to the start position, in a plane perpendicular to the fixed support.

A test shall be conducted at a rate of two cycles per minute. The time duration of one cycle shall approximate but not exceed 15 seconds. Continuity of all conductors shall be monitored continuously. The test shall be stopped and the number of cycles recorded In the event of failure of any conductor. The test shall consist of 3000 cycles of torsion. At the conclusion of this test, the cable shall be subjected to the test of 4.5.4.

4.5.10 Environmental tests.

4.5.10.1 Salt water immersion (Type I only). A length of completed cable shall be subjected to a 30-day immersion test. A cable shall be attached to a fixture so that the exposed ends will not be immersed in sea water. The cable shall be immersed for 16 hours and removed for a period of 8 hours. This sequence shall be repeated 30 times. No fresh water rinse shall be performed during the 30 cycles. At the conclusion of this test the cable shall be examined for conformance to the requirements of 3.6.1.

4.5.10.2 Hydrostatic pressure (Type II only). An externally sustained hydrostatic pressure of 1000 psi (7MPa) maintained for 90 minutes shall be applied to sample lengths of cable of not less than 7 feet (2.15 metres) or more than 9 feet (2.75 metres). Pressure shall be applied on the jacket in such a manner that the end faces of the cable can be examined for evidence of moisture leaks between the conductors.

Following the test, the jacket shall be stripped from the cable and the cable core examined thoroughly for evidence of moisture and for conformance to the requirements of 3,6.1.

Fifteen samples shall be taken from each lot, or two samples per 1000 feet (300m) for lots under 10,000 feet (3km), for hydrostatic testing.

4.5.10.3 Temperature and humidity. A 20-foot (7-metre) length of completed cable shall be placed in a test chamber such that the exposed ends are not inside the chamber and are prepared for monitoring the electrical characteristics. The cable shall then be subjected to a temperature of +122 degrees F (50° C) for 2 hours. During this 2-hour period the cable shall be subjected to the test of 4.5.4.

(a) After 2 hours, the cable shall be subjected to five 24-hour cycles of 16 hours at +122 degrees F (50° C) and 8 hours at +86 degrees F (30° C). The relative humidity shall be maintained constant at 95 percent during the five cycles. While at +122 degrees F (50° C), the test of 4.5.4 shall be performed.

(b) Upon completion of five cycles, the temperature shall be maintained at +122 degrees F (50° C) for 8 hours. During the 8-hour period, the relative humidity shall be 50 percent for the first 2 hours, 95 percent during the next 4 hours , and 40 percent during the final 2 hours. At 30-minute intervals the continuity of all conductors shall be checked. Upon completion, the test of 4.5.4 shall be performed.

NOTE: The transitions between temperatures shall be accomplished within the 8-hour period so that the time at the high temperature is approximately 16 hours. Each transition shall not exceed 1-1/2 hours.

4.5.10.4 Temperature and humidity (storage mode). The cable assembly, in the packaged condition, shall be subjected to a 28-day temperature and humidity test. The total teat is made up of two 14-day temperature and humidity cycles. The term “chamber temperature”, as used In this test, refers to the temperature of the air Immediately surrounding the equipment. Temperature may be changed(l) by varying the temperature of a single chamber or (2) by moving the test specimen from one constant temperature chamber to another. The fluctuations from all specified temperatures shall not exceed 10 degrees F (5.5° C). The following test sequence is based on a start of 0800 Monday. Regardless of the actual day and time the test is initiated, there shall be no deviation from the test sequence as prescribed.

(a) At 0800 Monday place the test specimen in a chamber which has been maintained at -65 degrees F (-54° C) for at least 2 hours immediately prior to 0800 (humidity not controlled).

(b) At 1600 Monday the chamber temperature shall be changed to +160 degrees F (71° C) and 95 +3 -O percent humidity. The chamber temperature shall reach +160 degrees F (71° C) and 95 percent humidity not later than 1800.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

(c) The test specimen shall be held under these conditions until 0800 Tuesday, at which time the temperature decrease shall begin. The rate of decrease of chamber temperature shall be equal to or greater than 36 degrees F (20° C) per hour for at least 2-1/2 hours. Thus, at 1030, the temperature of the air surrounding the test specimens shall be +70 degrees F (21° C) or lower. The chamber temperature shall reach -65 degrees F (-54° C) no later than 1400 and this temperature shall be held until 1600.

(d) At 1600 Tuesday the chamber temperature shall be changed to +160 degrees F (71° C) and 95 +3 -0 percent humidity as rapidly as practicable (not later than 1800) and held until 0800 Wednesday.

(e) On Wednesday, Thursday, and Friday the operations performed on Tuesday shall be repeated,

(f) The chamber temperature shall be raised to +160 degrees F (71° C) and 95 percent humidity on Friday evening and shall be so maintained until 0800 Monday.

(g) At 0800 Monday the sequence of operations described above for Tuesday of the first week shall be followed and shall be repeated daily until Friday of the second week.

(h) On Friday, the chamber temperature shall be reduced to -65 degrees F (-54 °C) and this temperature shall be maintained until 0800 Monday of the third week. The cycle will be complete at 0800 Monday.

4.5.10.4.1 The above sequence of temperature and humidity conditions will constitute one cycle. Two such cycles shall be applied. The second cycle shall be completed at 0800 Monday of the fifth week. At that time, the test specimen shall be allowed to return to room temperature. Without drying, wiping, or baking, the test specimen shall be subjected to the tests of 4.5.3 and 4.5.4.

4.5.10.5 Tropical environment (Fungus). For this test, the test specimen (a length of completed cable) shall be mounted in the test chamber in such a manner as to prevent exposing the cable ends to the environment.

(a) The test specimen shall be mounted as specified above in a mold chamber in which a temperature Of 86 degrees F ± 5 degrees F (30° ± 3° C) and a relative humidity of 90 percent ± 5 percent is maintained, The test specimen, while so housed, shall be sprayed with a suspension of mixed spores. The test specimen shall be maintained in this condition for a period of 28 days. At the end of this period, the test specimen shall be allowed to return to room temperature, shall be examined for evidence of being nutrient to fungi, shall be subjected to the tests of 4.5.3 and 4.5.4, and continuity of all conductors shall be determined. Color photographs shall be taken of any fungus growth on the cable.

(b) One type of fungus from each group listed in MIL-T-18404 shall be used in the test specimen inoculation. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at 77 degrees F (25° C)) shall be prepared by sterilization in convenient containers (approximately 100 milliliters (cm^3) each). Approximately 10 milliliters (cm^3) of the solution shall then be introduced directly into a stock culture and shaken vigorously in order that a well sporulated suspension will result without disturbing the agar.

(c) This process shall be repeated for each type fungus. The separate spore suspension from the various types of fungi shall then be mixed together in an atomizer to provide a composite suspension. Actively growing cultures in ripe fruiting condition shall be used for each inoculation and under no condition will the inoculum be kept for more than an 8-hour period.

4.6 Packaging, packing, and marking. Packaging, packing, and marking shall be examined for conformance to the requirements of Section 5.

5 PREPARATION FOR DELIVERY

5.1 Packaging, The cable shall be wound on reels or in coils as specified in MIL-C-9l5. Cable in lengths of 250 to 1000 feet shall be shipped on reels, whereas cable in lengths of 25 *to* 249 feet shall be packaged in coils.

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T07:18Z
Check the source to verify that this is the current version before use.

5.2 Packing

5.2.1 Reels. Cable wound on reels requires no further packing.

5.2.2 Coils. Cable in coil shall be packed for domestic shipment, if specified in 6.2, in accordance with MIL-C-915.

5.3 Marking. Each reel and coil of cable shall be marked for shipment in accordance with MIL-C-915, with the exception that color stripes shall be as follows for the particular year *of* manufacture. This *color* cycle shall repeat itself every fifth year.

Year of Manufacture	Identifying Color
1975	White
1976	Red
1977	Green
1978	Orange
1979	Blue

6 NOTES

6.1 Intended use. The electric cable described herein forms a part of the torpedo control cables (LS 162531, LD 162533, LD 162430 and DL2865325). These units complete the fire control circuit between the torpedo and the fire control console.

6,2 Ordering data.

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) Type cable required and applicable drawings (see 1.2 and 3.3.3),

(c) Preproduction sample length, if other than 100 feet (30 metres). See 4.3.2.1.

(d) Assigned activity for preproduction inspection.

(e) Lot acceptance sample, if different from 4.3.2.2.

(f) Disposition of samples and failures. See 4.4.3.

(g) Packaging and packing required.

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.4.2	Certified inspection data	

DIDs (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Metrication. This document has been annotated with the metric equivalents of quantities not expressed in the International System of Units. The equivalent units are in accordance with ASTM E 380-72 as adopted by the DOD on 29 November 1973, and are included for the purpose of encouraging metrication. Rather than exact equivalents, the metric quantities are generally given in rounded amounts within the tolerances specified herein.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS

Project Number:
1355-N354

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

NSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:18Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Naval Sea Systems Command
Engineering Support Division
ATTN: SEA 605
Department of the Navy
Washington, DC 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:18Z
Check the source to verify that this is the current version before use.